Racing-grade hydrodynamics for uncrewed maritime systems

Wyspa Technologies is a small US engineering firm that predicts, designs and builds fast boats for Grand Prix sailing. We apply our in-house hydrofoil, hull-performance and composite-structures tools to small, fast and efficient uncrewed surface vessels.

ENTITY	UEI	CAGE	BUSINESS TYPE
Wyspa Technologies LLC	NWJRCE69AWH9	Pending	Small business

Core competencies

- **Velocity & hydrofoil performance prediction.** In-house VPP with Delft-series resistance, ITTC-57 friction, lifting-line foils, and ventilation and cavitation limits. Validated against measured TP52 and Swan 42 data.
- **Composite laminate engineering.** Classical laminate theory with progressive ply failure (Tsai-Hill, Tsai-Wu), 550+ reinforcements and 90+ cores, and 3D-to-2D ply flattening with drape prediction and DXF cut files.
- **Foil flight dynamics & control.** Time-domain simulation of foil-borne craft in irregular seas with closed-loop ride-height, pitch and roll control.
- **Design-to-build.** NURBS hull modeling, Rhino and Grasshopper plugins, SolidWorks, and FARO 3D scanning for as-built verification.

Lead concept: autonomous foiling USV

GENERIC 7 M, 1,500 KG ELECTRIC USV	FOIL-BORNE	PLANING
Energy at 23 kn, calm	**1.13 kWh/nm**	**2.39 kWh/nm**
RMS vertical accel. SS2 / SS3 / SS4	**0.06/0.10/0.28 g**	**0.56/0.91/0.93 g**

Simulation estimates from Wyspa's time-domain model; not yet validated by tank or sea trials.

Differentiators

- Engineering proven in Grand Prix sailing, where small speed margins decide results.
- We own all core tools, so we can change them without licensing barriers and cycle quickly from analysis to design.
- Small team and fast iteration, suited to SBIR, BAA and OT timelines.

Past performance (commercial)

- 90+ yacht-performance and engineering projects, including TP52, Swan 42 and Etchells programs: VPP, rating-rule optimization, race-data analysis and weather routing.

NAICS

- 541330 Engineering Services
- 541715 R&D in Physical, Engineering & Life Sciences
- 336612 Boat Building
- 541511 Custom Computer Programming

Applications

- Harbor and port security · distributed logistics · ocean sensing
- Hydrodynamic and composite subcontracts for USV builders

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